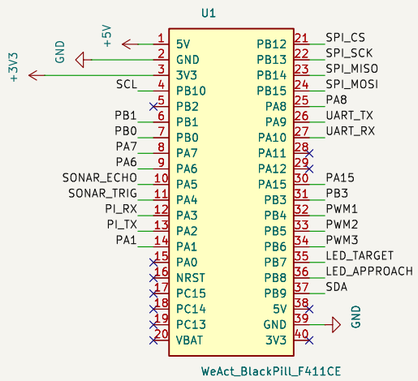
POWER: USB-C on the Black Pill feeds everything. +5V = USB 5V (no protection on the Black Pill),
+3V3 = Black Pill regulator (~300 mA total, MCU included). Do not also feed 5V from the Pi.

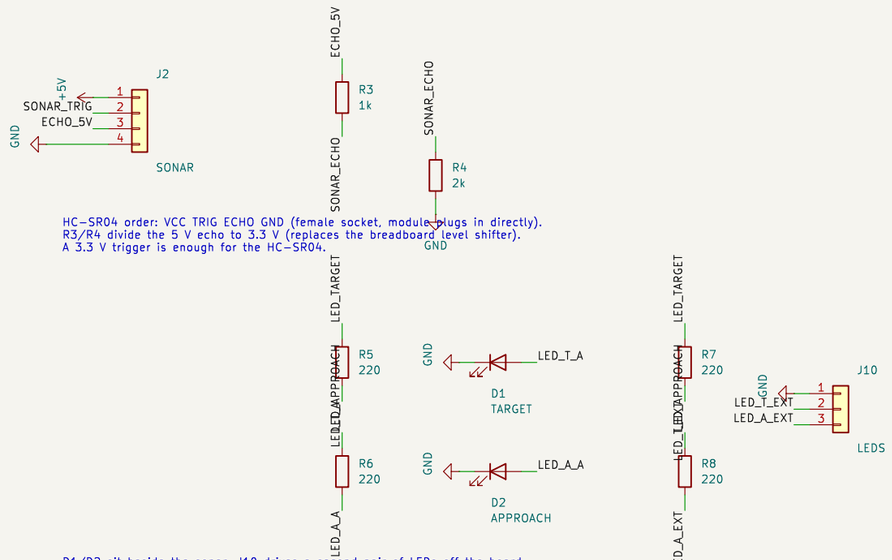
WeAct Black Pill V3.1 (STM32F411CEU6), socketed



Left free: PB2 (BOOT1), PA0 (KEY button), PA11/PA12 (USB),
PC13 (LED), PC14/PC15 (32 kHz crystal), NRST, VBAT.
Pins 38 (5V) and 40 (3V3) unused: the Black Pill ties them to pins 1 and 3.



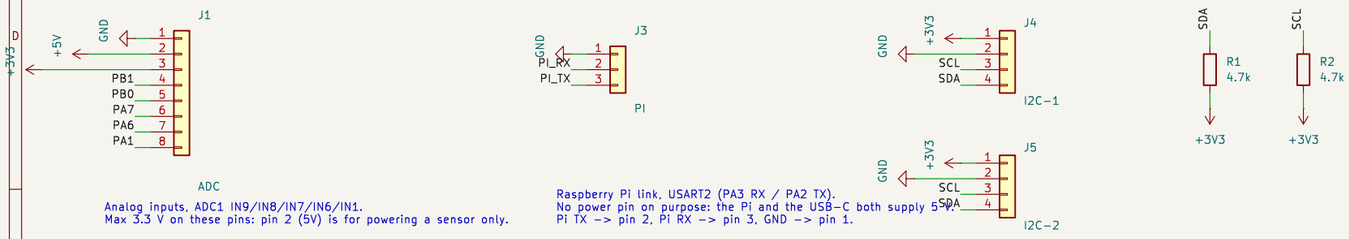
SONAR + GUIDE LEDS (right edge: the HC-SR04 plugs straight in and faces the user)



HC-SR04 order: VCC TRIG ECHO GND (female socket, module plugs in directly).
R3/R4 divide the 5 V echo to 3.3 V (replaces the breadboard level shifter).
A 3.3 V trigger is enough for the HC-SR04.

D1/D2 sit beside the sonar. J10 drives a second pair of LEDs off the board,
each through its own 220 R (R7/R8), so both pairs get full brightness.

TOP EDGE PORTS

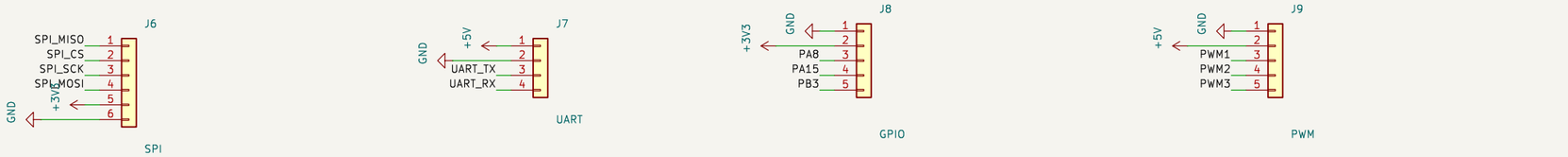


Analog inputs. ADC1 IN9/IN8/IN7/IN6/IN1.
Max 3.3 V on these pins: pin 2 (5V) is for powering a sensor only.

Raspberry Pi link, USART2 (PA3 RX / PA2 TX).
No power pin on purpose: the Pi and the USB-C both supply 5 V.
Pi TX -> pin 2, Pi RX -> pin 3, GND -> pin 1.

I2C2 on PB10 (SCL) / PB9 (SDA, AF9), both ports on one bus.
Leave R1/R2 off if the breakouts already have pull-ups.

BOTTOM EDGE PORTS (JST-XH, same pin orders as the ESP32 Home Node)



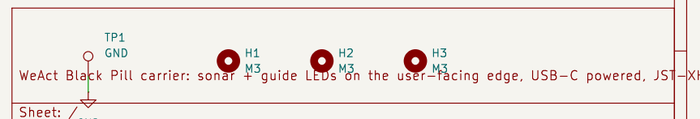
SPI2: MISO=PB14 CS=PB12 SCK=PB13 MOSI=PB15

USART1: TX=PA9 (pin 3), RX=PA10 (pin 4). Same pin numbers on every MODE port.

PA15/PB3 start up as JTAG pins: firmware must set SWD mode before using them.

PWM1/2 = PB4/PB5 (TIM3_CH1/CH2), PWM3 = PB6 (TIM4_CH1).
3.3 V logic. 5V pin is for powering the load.

MOUNTING / TEST



WeAct Black Pill carrier: sonar + guide LEDs on the user-facing edge, USB-C powered, JST-XH ports

Sheet: /		
File: stm32_carrier_board.kicad_sch		
Title: STM32 Home Node		
Size: A3	Date: 2026-10-24	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 1/1